Reciprocal Sums and the Sylvester Recurrence

Erdős Problem #243

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Abstract

We prove irrationality of the reciprocal sum when strictly increasing positive integers satisfy $a_n^2/a_{n+1} = 1 + 3/n + o(n^{-3})$. Rationality would give integer tail numerators whose fourth differences eventually vanish. The denominator recurrence then forces a square in a cubic field; trace identities and a four-term calculation modulo seven give a contradiction. Under rationality and the quadratic growth limit, we also give separate criteria for the eventual Sylvester recurrence using bounded numerator increases and weighted first crossings. The unrestricted Erdős–Graham question remains open.

1 Introduction

The Sylvester recurrence $a_{n+1} = a_n^2 - a_n + 1$ gives a rational reciprocal sum by telescoping:

$$\frac{1}{a_n - 1} = \frac{1}{a_n} + \frac{1}{a_{n+1} - 1}.$$

For an increasing sequence of positive integers that eventually satisfies this recurrence, the tail sum at every sufficiently large index is $1/(a_n - 1)$. Erdős and Graham asked whether every rational reciprocal sum with $a_{n+1} \sim a_n^2$ arises in this way.

Problem 1.1 (Erdős Problem 243)**.** Let $1 \le a_1 < a_2 < \cdots$ be a sequence of integers with

$$\lim_{n\to\infty} \frac{a_n}{a_{n-1}^2} = 1 \qquad \text{and} \qquad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then $a_n = a_{n-1}^2 - a_{n-1} + 1$ for all sufficiently large n.

Erdős and Graham [5, p. 64] asked this question; see also Erdős [6, p. 105] and Bloom's catalogue [11]. We prove the following irrationality result for a restricted class of sequences with $a_{n+1} \sim a_n^2$.

Theorem 1.2 (cubic-rate irrationality)**.** *A strictly increasing sequence of positive integers with*

$$a_n^2/a_{n+1} = 1 + \frac{3}{n} + o(n^{-3})$$

has irrational reciprocal sum.

Authorship and AI use. Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

The Sylvester recurrence gives $a_n^2/a_{n+1} - 1 = O(1/a_n)$, so the cubic-rate hypothesis excludes that behaviour. It imposes much more than the quadratic limit in Problem 1.1.

For example, set $a_1 = 8$ and

$$a_{n+1} = \left\lceil \frac{na_n^2}{n+3} \right\rceil.$$

This gives $8, 16, 103, 5305, \ldots$. From $a_{n+1} \geq a_n^2/4$ we obtain $a_n \geq 4 \cdot 2^{2^{n-1}}$ by induction. The error introduced by the ceiling is less than one, and hence

$$0 \leq 1 + \frac{3}{n} - \frac{a_n^2}{a_{n+1}} < \frac{16}{a_n^2} = o(n^{-3}).$$

The theorem applies to this explicitly defined sequence.

Koizumi's product criterion [10, Corollary 4] requires the increments of P_n/a_n, where $P_n = \prod_{j<n} a_j$, to have nonpositive upper limit. Corollary 4.9 below allows any finite upper limit. At the cubic rate, however, P_n/a_n grows like n^3 and its increments like n^2; neither criterion applies. This comparison does not decide whether the LCM criteria apply: those also depend on repeated factors in P_n.

Rationality supplies a positive integer tail numerator C_n, but not a bounded one: here it grows cubically. We therefore apply integrality to its finite differences. The exact comparison sequence $F_n = n(n+1)(n+2)$ has $F_{n+1}/F_n = 1 + 3/n$. The precision in the hypothesis forces $\Delta^4 C_n \to 0$, hence eventual equality $C_n = AF_n + B$. This extraction uses the numerator ratio and integrality, not the exact denominator recurrence.

The denominator recurrence now becomes decisive. It imposes a square condition at every root of a cubic modulo almost every prime. Chebotarev's theorem turns these residue conditions into a square in a cubic number field, where traces restrict the numerator to two polynomials. A four-term calculation modulo seven excludes both. Section 3 follows these two stages: first extract the polynomial, then test whether it can satisfy the exact recurrence.

Sections 4 and 5 give bounded-increment and summability criteria. Section 6 clears tails by a least common multiple and uses congruences to control first crossings of new heights; its convergence hypothesis is not derived from the quadratic limit. The companion paper also treats nonintegral rates and proves a stronger cubic disagreement theorem. Its proof locations and the formal sources are listed in Appendix B.

2 Tail numerators

For the Sylvester sequence $2, 3, 7, 43, \ldots$, whose reciprocal sum is 1, the successive tails are $1, 1/2, 1/6, 1/42, \ldots$. Multiplication by the products $1, 2, 6, 42, \ldots$ leaves the constant integer numerator 1. For any rational reciprocal sum, including a denominator of the whole sum in these products again gives integer numerators. We keep these fractions unreduced: the exact multiplicative denominator update will transfer the growth rate to the numerator.

Assume now that a_n are strictly increasing positive integers, $a_{n+1}/a_n^2 \to 1$, and $\sum_{n\geq 1} 1/a_n = p/q$ with positive integers p, q. All indices in this construction start at 1.

Put

$$x_n = \sum_{k\geq n} \frac{1}{a_k}, \qquad P_n = \prod_{1\leq j<n} a_j, \qquad D_n = qP_n, \qquad C_n = D_n x_n.$$

Here $P_1 = 1$, $D_1 = q$ and $C_1 = p$. Since every a_k with $k < n$ divides P_n, subtracting the finite prefix from p/q gives

$$C_n = pP_n - q \sum_{1\leq k<n} \frac{P_n}{a_k} \in \mathbb{N}_{>0}.$$

Positivity comes from $C_n = D_n x_n$. Removing $1/a_n$ from the tail gives

$$C_{n+1} = a_n C_n - D_n, \qquad D_{n+1} = a_n D_n. \tag{1}$$

Koizumi constructs these coordinates in Lemma 4 [10, pp. 11–12]. On a pseudo-greedy tail his c_n, d_n, e_n correspond to $C_n, D_n, D_n - (a_n - 1)C_n$. If the tail is restarted with its value in lowest terms, both coordinates are divided by the same fixed integer. Reducing at every step would instead introduce varying cancellation factors into the numerator ratio. We therefore extract the polynomial before dividing by the eventual fixed gcd in Section 3.2.

For large n, $a_{n+1} \geq a_n^2/2 \geq 2a_n$. The terms after $1/a_{n+1}$ sum to at most $2/a_{n+2} \leq 4/a_{n+1}^2$, so

$$x_n = \frac{1}{a_n} + \frac{1}{a_{n+1}} + O(a_{n+1}^{-2}), \qquad \frac{C_{n+1}}{C_n} = \frac{a_n^2}{a_{n+1}} + O(1/a_n). \tag{2}$$

For the second estimate, the first gives $a_n x_n = 1 + O(1/a_n)$. Use this in

$$\frac{C_{n+1}}{C_n} = \frac{a_n^2}{a_{n+1}} \frac{a_{n+1}x_{n+1}}{a_n x_n}.$$

The last factor is $1 + O(1/a_n)$, and $a_n^2/a_{n+1} \to 1$, giving the stated additive error. Quadratic growth makes $1/a_n = o(n^{-k})$ for every fixed k, so this error preserves the precision required in the cubic theorem. We keep the original index n throughout that proof. Although an eventual recurrence survives removal of a finite prefix, the expression $1 + 3/n + o(n^{-3})$ changes its lower-order terms under an index shift.

3 Proof of cubic-rate irrationality

Suppose, towards a contradiction, that the reciprocal sum in Theorem 1.2 is rational. Construct C_n, D_n as in Section 2. Equation (2) transfers the rate without changing its precision:

$$\frac{C_{n+1}}{C_n} = 1 + \frac{3}{n} + o(n^{-3}). \tag{3}$$

3.1 An eventual cubic polynomial

We use $F_n = n(n+1)(n+2)$ rather than n^3: its successive ratio is *exactly* $1 + 3/n$, so dividing by it removes the model without introducing lower-order errors. Write (3) as $C_{n+1} = (1 + 3/n + \varepsilon_n)C_n$, where $\varepsilon_n = o(n^{-3})$. Dividing out the exact model gives

$$z_n = \frac{C_n}{F_n}, \qquad \frac{z_{n+1}}{z_n} = 1 + \frac{\varepsilon_n}{1 + 3/n}.$$

The logarithmic increments are $o(n^{-3})$, so their absolute tails are $o(n^{-2})$. Hence $z_n \to A > 0$ with $z_n - A = o(n^{-2})$. Multiplying by F_n gives

$$C_n = AF_n + \delta_n, \qquad \delta_n = o(n).$$

An $o(n)$ error can still oscillate too much for finite differences. To control its first difference, subtract the numerator update for C_n and the comparison update for AF_n:

$$\Delta\delta_n = \frac{3}{n}\delta_n + \varepsilon_n C_n = o(1), \qquad \Delta h_n = h_{n+1} - h_n.$$

Here $C_n = O(n^3)$ makes the second term $o(1)$. Three more differences still tend to zero and kill the cubic model:

$$\Delta^4 C_n = \Delta^3(\Delta\delta_n) \longrightarrow 0.$$

Integrality turns this limit into eventual vanishing. Newton interpolation then expresses C_n as a rational polynomial of degree at most three. Since its difference from AF_n is $o(n)$, the cubic, quadratic and linear coefficients agree; only a constant can remain:

$$C_n = An(n+1)(n+2) + B, \qquad A \in \mathbb{Q}_{>0}, \quad B \in \mathbb{Q}. \tag{4}$$

The precision matters at the subtraction step. The integer sequence $F_n + (-1)^n$ has ratio $1 + 3/n + O(n^{-3})$ and never agrees eventually with a polynomial; its error has an oscillating first difference. This example concerns polynomial extraction alone. It is not asserted to satisfy the denominator recurrence.

3.2 Reduction and the constant term

We now use the denominator recurrence. Both updates preserve common divisors, so $G_n = \gcd(C_n, D_n)$ divides G_{n+1} and every later numerator. On the polynomial tail it therefore divides the third difference $6A$, a fixed positive integer. The fourth difference gave polynomiality; the nonzero third difference now bounds cancellation. Thus G_n stabilises at a positive integer g.

Dividing by this fixed g gives $u_n = C_n/g$, $v_n = D_n/g$ in lowest terms without changing the recurrences. Their coprimality persists as follows.

Proposition 3.1 (persistent coprimality)**.** *Let a_n, u_n, v_n be integer sequences with $u_n > 0$, $v_n \geq 0$ and $\gcd(u_n, v_n) = 1$ satisfying*

$$u_{n+1} = a_n u_n - v_n, \qquad v_{n+1} = a_n v_n.$$

Then $\gcd(a_n, v_n) = 1$. The a_n are pairwise coprime, and $\gcd(a_i, u_t) = 1$ whenever $i < t$.

Proof. A prime dividing both a_n and v_n would divide u_{n+1} and v_{n+1}, contrary to reducedness. For $i < t$, the factor a_i divides v_t. Reducedness gives $\gcd(a_i, u_t) = 1$, and $\gcd(a_t, v_t) = 1$ gives $\gcd(a_i, a_t) = 1$. □

In the cubic case, adjacent numerators are also coprime, since $\gcd(u_n, u_{n+1}) = \gcd(u_n, v_n) = 1$. Write the polynomial for u_n as

$$Q(n) = \frac{m}{6}n(n+1)(n+2) + c. \tag{5}$$

The third difference $m = 6A/g$ is a positive integer. Since $n(n+1)(n+2)/6 = \binom{n+2}{3}$ is an integer, any sufficiently late value $u_n = Q(n)$ gives

$$c = u_n - m\binom{n+2}{3} \in \mathbb{Z}.$$

Adjacent coprimality now restricts this integer c. If a prime ℓ divides c, choose an arbitrarily late $n \equiv -1 \pmod{6\ell}$. Both three-factor products contain $n+1$, so division by 6 leaves a factor ℓ, even for $\ell = 2, 3$. Thus ℓ divides both $Q(n)$ and $Q(n+1)$, contradicting adjacent coprimality. This also excludes $c = 0$. Thus

$$m \in \mathbb{N}_{>0}, \qquad c \in \{1, -1\}.$$

Irreducibility follows from persistence of denominator primes. The pairwise coprime multipliers $a_n > 1$ supply infinitely many distinct primes; each divides every later v_t and no later u_t. A rational root of Q would reduce to a root modulo all but finitely many primes. Choose a persistent prime outside those exceptions. Every sufficiently late index in the root's residue class would give $\ell \mid Q(t) = u_t$, contradicting reducedness. Thus Q has no rational root and, being cubic, is irreducible.

Set $\kappa = m/6$, $\eta = 6c/m$, and

$$f(T) = T^3 - T + \eta, \qquad Q(n) = \kappa f(n+1).$$

3.3 A square forced by a zero numerator

At a zero of the numerator polynomial modulo a prime, the recurrence must still relate the two neighbouring nonzero numerators. Eliminating v_n from two exact updates makes this constraint visible:

$$u_{n+2} = (a_n + a_{n+1})u_{n+1} - a_n^2 u_n. \tag{6}$$

Whenever $u_k \equiv 0 \pmod{\ell}$, it follows that

$$-u_{k-1}u_{k+1} \equiv (a_{k-1}u_{k-1})^2 \pmod{\ell}.$$

Thus the negative product of the neighbouring numerators must be a square. Since our numerator is a polynomial, its roots modulo ℓ tell us exactly where to apply this observation.

Let $\ell \nmid 6m$ be prime and let $r \in \mathbb{F}_\ell$ satisfy $f(r) = 0$. The constant $\eta = 6c/m$ is nonzero modulo ℓ, so $r \notin \{0, 1, -1\}$. Choose a sufficiently late k with $k + 1 \equiv r \pmod{\ell}$. Then $u_k = \kappa f(k+1) \equiv 0$, whereas

$$f(r-1) = -3r(r-1), \qquad f(r+1) = 3r(r+1).$$

Consequently, in $\mathbb{F}_\ell$,

$$(a_{k-1}u_{k-1})^2 = -u_{k-1}u_{k+1} = 9\kappa^2 r^2(r^2 - 1).$$

Dividing by the nonzero square $9\kappa^2 r^2$ gives

$$\ell \nmid 6m, \quad r \in \mathbb{F}_\ell, \quad f(r) = 0 \quad \Longrightarrow \quad r^2 - 1 \in (\mathbb{F}_\ell^\times)^2 \qquad (\ell \text{ prime}). \tag{7}$$

The condition holds for *every* root; primes with no root impose no condition. This quantifier matters in passing to a number field.

Let α be a root of f and put $K = \mathbb{Q}(\alpha)$, a cubic field. Suppose $\alpha^2 - 1$ is a nonsquare in K, and choose β with $\beta^2 = \alpha^2 - 1$. To contradict the modular condition, we seek a Frobenius that fixes α but exchanges β and $-\beta$. The nontrivial automorphism of $K(\beta)/K$ does exactly this. Extend it to $\sigma \in \mathrm{Gal}(M/\mathbb{Q})$, where M is a Galois closure of $K(\beta)/\mathbb{Q}$.

Chebotarev's density theorem [9, §3, author-version p. 15] supplies infinitely many unramified rational primes with Frobenius conjugate to σ; choosing a prime of M above each makes it equal to σ. Omit residue characteristic two, primes dividing $6m$, and the finitely many primes where α or β is nonintegral or β reduces to zero. Frobenius acts by the ℓ-th power on the residue field, so

$$\bar{\alpha}^{\ell} = \bar{\alpha}, \qquad \bar{\beta}^{\ell} = -\bar{\beta} \neq \bar{\beta}.$$

Thus $r = \bar{\alpha}$ lies in $\mathbb{F}_\ell$ and satisfies $f(r) = 0$, but neither root $\pm\bar{\beta}$ of $X^2 - (r^2 - 1)$ lies in that field. The condition for *every* root in (7) applies to this r, giving the contradiction. Hence some $\beta \in K$ satisfies $\beta^2 = \alpha^2 - 1$.

3.4 Traces in the cubic field

We now need the square relation to constrain the coefficient η. Factoring it produces reciprocal elements, avoiding a separate expansion of the square root in a basis of K:

$$(\alpha + \beta)(\alpha - \beta) = 1.$$

Put $z = \alpha + \beta$. Then $z^{-1} = \alpha - \beta$ and $2\alpha = z + z^{-1}$. Since $z \in K$ and $\alpha \in \mathbb{Q}(z)$, we have $\mathbb{Q}(z) = K$. Write the minimal polynomial of z as

$$p(T) = T^3 + bT^2 + dT + w, \qquad b, d, w \in \mathbb{Q}, \quad w \neq 0.$$

The minimal polynomial of z^{-1} is the monic reciprocal polynomial

$$\frac{T^3 p(T^{-1})}{w} = T^3 + \frac{d}{w}T^2 + \frac{b}{w}T + \frac{1}{w}.$$

We compute the same three traces from f and from the reciprocal pair z, z^{-1}. All traces are from K to $\mathbb{Q}$. The cubic for α gives:

$$\mathrm{Tr}(\alpha) = 0, \qquad \mathrm{Tr}(\alpha^2) = 2, \qquad \mathrm{Tr}(\alpha^3) = -3\eta.$$

Newton's identities for p and its reciprocal give

$$d = -bw, \qquad b^2 + b(w - w^{-1}) = 1, \qquad \eta = \frac{(b^2 + 1)(w + w^{-1})}{8}. \tag{8}$$

For completeness, the first equality follows from $0 = 2\,\mathrm{Tr}(\alpha) = -b - d/w$. After substituting $d = -bw$, Newton's identities give

$$\mathrm{Tr}(z^2) + \mathrm{Tr}(z^{-2}) = 2b^2 + 2b(w - w^{-1}),$$
$$\mathrm{Tr}(z^3) + \mathrm{Tr}(z^{-3}) = -3(b^2 + 1)(w + w^{-1}).$$

The first line is $4\,\mathrm{Tr}(\alpha^2) - 6 = 2$. The second is $8\,\mathrm{Tr}(\alpha^3) = -24\eta$, because the mixed terms in $(z + z^{-1})^3$ have trace $3\,\mathrm{Tr}(z + z^{-1}) = 0$. These give the other two equalities in (8).

Multiplying the middle equation by w and bringing all terms to one side gives $(bw - 1)(b + w) = 0$. Write $w = r/s$ in lowest terms, with $r \neq 0$, $s > 0$. Substituting $b = -w$ or $b = 1/w$ into the last equation and using $\eta = 6c/m$ gives respectively

$$m = \frac{48c\,rs^3}{(r^2 + s^2)^2} \qquad \text{or} \qquad m = \frac{48c\,r^3 s}{(r^2 + s^2)^2}.$$

Here the integer scale m is decisive: coprimality $\gcd(r^2+s^2, rs) = 1$ prevents cancellation with the variable factors in either numerator, so $(r^2 + s^2)^2$ divides 48. Hence $r^2 + s^2$ is 1, 2 or 4. Both r and s are nonzero, which leaves only 2: a sum of two nonzero integer squares cannot equal 1 or 4. Thus $|r| = s = 1$, and positivity of m gives $m = 12$. The only remaining numerators are

$$Q(n) = 2n(n + 1)(n + 2) + 1 \quad \text{and} \quad Q(n) = 2n(n + 1)(n + 2) - 1.$$

3.5 The contradiction modulo seven

The three-term square condition has reduced the possibilities to two cubics. We now use one more term: the denominator at the middle zero must also come from the preceding exact update. In the plus case, choose a late block starting at $n \equiv 0 \pmod 7$; in the minus case, start at $n \equiv 1 \pmod 7$. The four successive numerators have residues

$$(u_n, u_{n+1}, u_{n+2}, u_{n+3}) \equiv \begin{cases} (1, 6, 0, 2) & (c = 1), \\ (4, 5, 0, 1) & (c = -1). \end{cases}$$

Write d_0, d_1 for the residues of v_n, v_{n+1} and $c_0, c_1, 0, c_3$ for the displayed numerator residues. From $0 = a_{n+1}c_1 - d_1$ and $c_3 = -a_{n+1}d_1$ we get $d_1^2 = -c_1c_3 = 2$. Thus $d_1 \in \{3, 4\}$. The first update, however, gives

$$d_1 = a_n d_0 = \frac{d_0(d_0 + c_1)}{c_0}.$$

Completing the square in these two cases gives, in $\mathbb{F}_7$,

$$d_1 + 2 = \begin{cases} (d_0 + 3)^2 & (c = 1), \\ 2(d_0 - 1)^2 & (c = -1). \end{cases}$$

Since $2 = 3^2$ in this field, $d_1 + 2$ must be a square in either case. But $d_1 \in \{3, 4\}$ would give $d_1 + 2 \in \{5, 6\}$, both nonsquares. The middle three terms admit $d_1 = 3$ or 4; neither extends backwards through the preceding exact update. Both cubics are excluded, proving Theorem 1.2.

Section 2 of the companion proves positive lower density of disagreement with each rational cubic of this form. That lower bound depends on the orbit and the cubic. For Theorem 1.2, exclusion of eventual equality suffices.

4 Bounded increases

We return to the rational tails of Section 2. The Sylvester example had a constant integer numerator. To measure departure from this behaviour, put

$$E_n = D_n - (a_n - 1)C_n = C_n - C_{n+1}.$$

Negative errors are rises of C_n; zero error gives the tail $1/(a_n - 1)$. Instead of prescribing a polynomial, we now bound the rises. Quadratic growth makes all steps small relative to C_n; the additional absolute bound applies only to rises, not decreases.

Theorem 4.1 (bounded negative part)**.** *Let $a, C, D : \mathbb{N} \to \mathbb{N}$ satisfy $a_n > 1$, $C_n > 0$, and*

$$C_{n+1} + D_n = a_n C_n, \qquad D_{n+1} = a_n D_n \quad (n \geq 0).$$

Put $E_n = D_n - (a_n - 1)C_n$. If

$$\frac{E_n}{C_n} \longrightarrow 0 \quad \textit{and} \quad E_n \geq -B \quad \textit{for all sufficiently large } n$$

for some integer $B \geq 0$, then $E_n = 0$ for all sufficiently large n.

The linked theorem uses the equivalent division-free condition: for every $K \in \mathbb{N}$, eventually $K|E_n| < C_n$. Under this condition, absorption either settles the theorem or leaves $C_n \to \infty$. A stable gcd and a forbidden block will exclude the latter alternative.

4.1 Zero errors and descent

Write $z_+ = \max(z, 0)$. The identities below use (1); positivity and $|E_n| < C_n$ are imposed only where stated.

Proposition 4.2 (error identities)**.** *For integer sequences satisfying (1),*

$$C_{n+1} = C_n - E_n, \qquad (a_{n+1} - a_n^2 + a_n - 1)C_{n+1} = a_n^2 E_n - E_{n+1}.$$

Proof. The first identity follows by substituting $D_n = E_n + (a_n - 1)C_n$ into the numerator recurrence. Substituting this expression at the next index and using $D_{n+1} = a_n D_n$ gives the second. □

Equation (2) now gives $E_n/C_n = 1 - C_{n+1}/C_n \to 0$, so $|E_n| < C_n$ for all sufficiently large n. Koizumi proves this relative-gap limit for an eventual pseudo-greedy tail [10, Corollary 3, p. 9]. His rounding convention also gives $-C_n/2 \leq E_n < C_n/2$, a stronger bound than we need.

Koizumi's Lemma 3 gives zero absorption for pseudo-greedy tails, and Proposition 1(2) uses descent when their gaps are eventually nonnegative [10, p. 10 and p. 14]. The following form separates what the two arguments require.

Theorem 4.3 (absorption and descent)**.** *Let $a, C, D : \mathbb{N} \to \mathbb{N}$ satisfy $a_n > 1$, $C_n > 0$ and (1). If $|E_n| < C_n$ for every n, then $E_n = 0$ implies $E_{n+1} = 0$. For any positive integer sequence C_n with $C_{n+1} = C_n - E_n$, eventual nonnegativity of E_n implies its eventual vanishing.*

Proof. When $E_n = 0$, the second identity in Proposition 4.2 shows that C_{n+1} divides E_{n+1}. The inequality $|E_{n+1}| < C_{n+1}$ then forces $E_{n+1} = 0$. For the second assertion, $C_{n+1} = C_n - E_n$ makes the positive integer sequence C_n eventually nonincreasing, hence constant. □

Corollary 4.4 (two zero errors)**.** *For integer sequences satisfying* (1)*, if $E_n = E_{n+1} = 0$ and $C_{n+1} \neq 0$, then $a_{n+1} = a_n^2 - a_n + 1$. In particular, if $a, C, D : \mathbb{N} \to \mathbb{N}$ also satisfy $a_n > 1$ and $C_n > 0$, eventual zero error implies the eventual Sylvester recurrence.*

Proof. Cancel the nonzero factor C_{n+1} in the second error identity. □

Thus a late zero is absorbing. Without late zeros, $|E_n| \geq 1$ and $|E_n|/C_n \to 0$ force $C_n \to \infty$. The scalar update permits $C_n = n + 1$, $E_n = -1$ forever; excluding this behaviour requires the denominator recurrence.

4.2 A stable gcd and a forbidden block

Proposition 4.5 (gcd stabilisation)**.** *Let $a, C, D : \mathbb{N} \to \mathbb{N}$ satisfy $a_n > 1$, $C_n > 0$ and* (1)*. If some fixed integer $B \geq 1$ satisfies $-B \leq E_n < 0$ at infinitely many indices, then $G_n = \gcd(C_n, D_n)$ is eventually constant. Dividing C_n, D_n by its eventual value gives coprime sequences satisfying the same recurrences.*

Proof. We have $G_n \mid G_{n+1}$ and $G_n \mid E_n$. Given n, choose $t \geq n$ with $-B \leq E_t < 0$. Then $G_n \leq G_t \leq -E_t \leq B$. Hence the positive divisibility chain stabilises. Division by its stable value preserves the recurrences and gives coprime numerators and denominators. □

There are two distinct uses of the bound. Arbitrarily late negative errors in a fixed interval suffice to stabilise the gcd. The crossing argument below needs *every* sufficiently late upward step bounded; otherwise a single large rise could jump over the forbidden block.

A sequence avoiding multiples of 2 and 3 cannot cross the pair $6k + 2, 6k + 3$ from below with upward steps at most two: its first value at least $6k + 2$ would be one of those two integers. Decreases before that crossing do not help. For a general step bound B, the Chinese remainder theorem supplies a forbidden block of length B.

Lemma 4.6 (consecutive multiples)**.** *For pairwise coprime integers $m_0, \ldots, m_{B-1} \geq 2$ and every lower bound, there is a larger t such that $m_i \mid t + i$ for each $i < B$.*

Proof. Solve $t \equiv -i \pmod{m_i}$ by the Chinese remainder theorem and add multiples of $\prod_i m_i$ to exceed the prescribed bound. □

Theorem 4.7 (Chinese remainder theorem and first crossing)**.** *Let $u : \mathbb{N} \to \mathbb{N}$ tend to infinity and let $B \geq 1$ be an integer with $u_{n+1} \leq u_n + B$ for every n. There is no sequence of pairwise coprime integers $m_i \geq 2$ for which $\gcd(m_i, u_t) = 1$ whenever $i < t$.*

Proof. Apply Lemma 4.6 to $m_0, \ldots, m_{B-1}$, choosing $t > \max(u_0, \ldots, u_B)$. There is a first $n > B$ with $u_n \geq t$. Then

$$t \leq u_n \leq u_{n-1} + B < t + B.$$

Thus $u_n = t + i$ for some $i < B < n$, and $m_i \mid u_n$, contrary to $\gcd(m_i, u_n) = 1$. □

Bado proves a two-sided bounded-error criterion by this forbidden-block argument [7, Theorem 5.1, pp. 4–5]. The gcd stabilisation above allows us to use only the lower error bound.

Proof of Theorem 4.1. Suppose the conclusion fails. Absorption excludes late zeros; $|E_n| \geq 1$ and the relative limit then give $C_n \to \infty$. Integer descent forces infinitely many late negative errors, where $1 \leq -E_n \leq B$. Thus $B \geq 1$ and Proposition 4.5 gives a stable gcd g. Restrict to a tail on which the gcd is g and $E_n \geq -B$, then reindex that tail. Dividing by g gives $u_n = C_n/g \to \infty$ and

$$u_{n+1} - u_n = -E_n/g \leq B.$$

Proposition 3.1 makes the multipliers $a_i \geq 2$ pairwise coprime and coprime to every later numerator, contradicting Theorem 4.7. □

Corollary 4.8. *Under Theorem 4.1, the multipliers satisfy $a_{n+1} = a_n^2 - a_n + 1$ eventually.*

Proof. Theorem 4.1 gives eventual zero error; positivity of C_n and Corollary 4.4 give the recurrence. □

4.3 The condition in the original sequence

Let $\gamma_n = a_n^2/a_{n+1} - 1$. For $t_n = P_n/a_n$,

$$t_{n+1} - t_n = t_n\gamma_n.$$

We bound upward increments, not values or decreases. The relation $C_n \sim qt_n$ does not control increments; instead an exact identity will give $E_n > -qt_n\gamma_n$, turning an upper increment bound into a lower error bound.

Corollary 4.9 (bounded increments of P_n/a_n)**.** *Let $a_1 < a_2 < \cdots$ be positive integers, $a_{n+1}/a_n^2 \to 1$, and $\sum_{n\geq 1} 1/a_n \in \mathbb{Q}$. Put $P_n = \prod_{j<n} a_j$. If*

$$\limsup_{n\to\infty} \frac{P_n}{a_n}\left(\frac{a_n^2}{a_{n+1}} - 1\right) < +\infty,$$

then $a_{n+1} = a_n^2 - a_n + 1$ for all sufficiently large n.

Proof. For the canonical integer tail, direct substitution gives

$$E_n + q\frac{P_n}{a_n}\gamma_n = qP_n\left(\frac{1}{a_{n+1}} - (a_n - 1)\sum_{k\geq n+2}\frac{1}{a_k}\right). \tag{9}$$

The tail estimate bounds the subtracted term by $4(a_n - 1)/a_{n+1}^2 < 1/a_{n+1}$ eventually. The positive remainder gives $E_n > -qt_n\gamma_n$, so the assumed upper bound bounds E_n below. Only the remainder's sign is needed here. The canonical tails satisfy the exact recurrences and, by (2), $E_n/C_n \to 0$. Theorem 4.1 and Corollary 4.8 finish. □

Koizumi's proof of Corollary 4(1) uses this product-weighted comparison [10, (11)–(12), p. 15]; that corollary assumes a nonpositive upper limit, whereas the argument here allows any finite upper limit. A corresponding LCM comparison appears below. For $a_n = b^{2^{n-1}}$, $b \geq 2$, the increments of P_n/a_n are zero. Since the sequence does not eventually satisfy the Sylvester recurrence, the corollary gives irrationality. For a Sylvester sequence starting at $a_1 > 1$, $a_n - 1 = (a_1 - 1)P_n$ and the increments tend to zero.

Dividing by n removes the nonsummable $1/n$ term. A summable upper error then suffices, without a lower bound on γ_n.

Corollary 4.10 (an inclusive one-sided $1/n$ bound)**.** *Let $a_1 < a_2 < \cdots$ be positive integers with $a_{n+1}/a_n^2 \to 1$ and $\sum_n 1/a_n \in \mathbb{Q}$. Suppose that for some $K \geq 0$ and $\varepsilon > 0$,*

$$\gamma_n := a_n^2/a_{n+1} - 1 \leq \frac{1}{n} + \frac{K}{n^{1+\varepsilon}} \quad \text{for all large } n.$$

Then the sequence is eventually Sylvester. In particular, the conclusion holds under the pointwise eventual bound $\gamma_n \leq 1/n$.

Proof. The exact model n has successive ratio $1 + 1/n$. Thus $(t_{n+1}/(n+1))/(t_n/n) = (1+\gamma_n)/(1+1/n) \leq 1 + K/[n^{\varepsilon}(n+1)]$. The convergent product of these upper bounds bounds t_n/n above; no positive limit is needed. Hence $t_n = O(n)$ and $t_n\gamma_n \leq t_n(\gamma_n)_+ = O(1)$, so Corollary 4.9 applies. □

The weaker condition $\limsup n(\gamma_n)_+ \leq 1$ does not suffice for the estimate $t_n = O(n)$. For example, the scalar ratios $1+\gamma_n = 1+(1+1/\log n)/n$ give a product of order $n \log n$. This scalar product refutes the intermediate linear estimate under the weaker rate, not the Sylvester conclusion for reciprocal tails.

For positive limiting increments, take $a_1 = 4$ and $a_{n+1} = \lceil na_n^2/(n+1) \rceil$, beginning $4, 8, 43, 1387$. Here $a_n \geq 2 \cdot 2^{2^{n-1}}$ and

$$0 \leq 1 + \frac{1}{n} - \frac{a_n^2}{a_{n+1}} < \frac{4}{a_n^2}.$$

Consequently $t_n \sim Kn$ and $t_{n+1} - t_n \to K > 0$. The corollary proves irrationality because $\gamma_n \sim 1/n$ is incompatible with the eventual Sylvester recurrence.

For the introduction's $3/n$ ceiling example, division by $n(n+1)(n+2)$ and summability of the rounding errors give $t_n \sim K_3 n^3$ and $t_{n+1} - t_n \sim 3K_3 n^2$, with $K_3 > 0$. Its irrationality follows from Theorem 1.2, beyond the bounded-increase criterion.

5 Finite total relative increase

Summable relative increases first bound C_n by an infinite product, then allow only finitely many rises; integer descent finishes. No denominator recurrence is needed. Relative increases tending merely to zero still allow $C_n = n + 1$ or $C_n = (n+1)^2$.

Theorem 5.1 (a convergent sum of relative increases)**.** *Let C_n be positive integers and E_n integers satisfying $C_{n+1} = C_n - E_n$. If*

$$\sum_n \frac{(-E_n)_+}{C_n} < \infty,$$

then $E_n = 0$ eventually. Neither denominator dynamics nor vanishing relative error is required.

Proof. Set $\delta_n = (-E_n)_+/C_n$. Then

$$C_N \le C_0 \prod_{n<N} (1+\delta_n) \le C_0 \exp\Bigg(\sum_n \delta_n\Bigg).$$

Choose an integer K above this bound. A strict rise of the integer sequence has size at least one, so it contributes at least $1/K$ to $\sum \delta_n$. Hence only finitely many rises occur. Thereafter C_n is a nonincreasing positive integer sequence, and its eventual constancy gives $E_n = 0$. □

The proof also allows positive real C_n with integer E_n: the values lie in $C_0 + \mathbb{Z}$, whose bounded positive part is finite. Discrete increments are essential to this argument. With real errors, $C_n = 1+1/(n+1)$ decreases forever with zero relative-increase sum; without positivity, $C_n = -n-1$, $E_n = 1$ does so too. The companion's Section 13 gives the details.

On an exact reciprocal-tail orbit the conclusion gives the Sylvester recurrence. For the gap sequence of the pseudo-greedy expansion, the same criterion, with the same product bound and integer descent, appears in the Erdős Problem a Day working report on Problem #243, dated 12 August 2026 [8, "A global termination criterion"]; the statement above is for any positive integer sequence with $C_{n+1} = C_n - E_n$. This summability condition and the bound in Theorem 4.1 are different estimates. For exact reciprocal tails with $|E_n|/C_n \to 0$, each is equivalent to eventual vanishing of E_n; neither estimate has been derived here from the unrestricted problem.

6 New maxima of an LCM numerator

The least common multiple retains only the largest power of each prime among $q, a_1, \dots, a_{n-1}$. It preserves integrality, not coprimality: a numerator may now take values forbidden in Section 4.2. The recurrence will instead forbid short positive jumps from selected multiples. Put

$$L_n = \operatorname{lcm}(q, a_1, \dots, a_{n-1}), \quad M_n = D_n/L_n, \quad U_n = L_n x_n = C_n/M_n, \quad V_n = E_n/M_n.$$

The denominator of x_n divides L_n, so $U_n \in \mathbb{Z}$ and $V_n = L_n - (a_n - 1)U_n \in \mathbb{Z}$. The fraction U_n/L_n need not be reduced. With $\rho_n = \gcd(L_n, a_n)$, the exact updates are

$$M_{n+1} = M_n \rho_n, \qquad \rho_n U_{n+1} = U_n - V_n, \qquad V_n = L_n - (a_n - 1)U_n.$$

These coordinates and updates also appear in Bado's September 2026 preprint. With his denominator parameter equal to q, his M_{n-1}, Δ_{n-1}, K_{n-1}, u_n and g_n are respectively our L_n, M_n, U_n, V_n and ρ_n; his (19) is the middle update above [7, Proposition 7.1 and (16)–(20), pp. 6–7].

Since $L_n \mid qP_n$ and $V_n = (L_n/(qP_n))E_n$, multiplying (9) by $L_n/(qP_n)$ gives, eventually,

$$0 < V_n + \frac{L_n}{a_n}\left(\frac{a_n^2}{a_{n+1}} - 1\right) \le \frac{L_n}{a_{n+1}}. \tag{10}$$

The lower bound transfers an upper bound on the growth defect to a lower bound on V_n. The upper bound will let us compare weighted series with a summable error.

Write $R_n = \max_{j\le n} U_j$ and $\mathcal{R} = \{n : U_{n+1} > R_n\}$. Since $V_n/U_n = E_n/C_n \to 0$, eventually $|V_n| < U_n$ and $-U_n \le 2V_n$. When $\rho_n \ge 2$, the first bound gives $U_{n+1} < U_n$; the second sharpens this to $U_{n+1} \le 3U_n/4$. Thus every late rise, in particular every record, has $\rho_n = 1$: its multiplier is coprime to L_n. At such a record write the full jump as $d_n = U_{n+1} - U_n = -V_n > 0$; at a contracting step that equality need not hold.

Theorem 6.1 (a convergent weighted sum over new maxima)**.** *Assume the growth and rationality hypotheses of Problem 1.1. Let $f : [1,\infty) \to [0,\infty)$ be finite and nonincreasing, with $\int_1^\infty f(t)\,dt = \infty$. Then the sequence is eventually Sylvester if and only if, for some integer $B \ge 0$,*

$$\sum_{n\in\mathcal{R}} (-V_n - B)_+ f(U_n) < \infty.$$

At a late record, the summand is $(d_n - B)_+ f(U_n)$, where $d_n = U_{n+1} - U_n$.

The full jump matters: with previous record 10, a step from 5 to 12 first crosses 11. Its jump is 7, but the record increases by only 2. This illustrates the two increments, not a denominator sequence satisfying (1).

For $B = 0$, the integral comparison

$$\int_{R_n}^{R_{n+1}} f(t)\,dt \le (R_{n+1} - R_n) f(U_n) \le d_n f(U_n)$$

already charges for every newly reached height. The arithmetic is needed to retain divergence after subtracting a fixed B at *each* record. We shall find a progression of heights whose first crossings still cost at least one unit of excess jump.

The weights $1/t$ and $1/[t\log(et)]$ are allowed; $1/t^2$ is not. One admissible weight and one fixed B suffice; B cannot grow with the record. The limit $V_n/U_n \to 0$ alone gives no summability.

Proof. Suppose the sequence is not eventually Sylvester. Absorption excludes late zeros of E_n, so $V_n = E_n/M_n$ is eventually a nonzero integer. Hence

$$\frac{1}{U_n} \le \frac{|V_n|}{U_n} = \frac{|E_n|}{C_n} \longrightarrow 0.$$

Thus $U_n \to \infty$. Infinitely many late records supply pairwise coprime multipliers: each is coprime to its L_n, which is divisible by every earlier multiplier. Since $a_n \to \infty$, for any fixed $B \ge 1$ choose B of them, $m_0, \dots, m_{B-1} > B$. Take T beyond their indices and the threshold for $\rho_n = 1$ at records.

Put $P = \prod_i m_i$ and choose x with $m_i \mid x + i$ by the Chinese remainder theorem. We select the heights $\tau = x + B + kP > R_T$, $k \in \mathbb{Z}$. They are spaced P apart.

At the first crossing of a selected height, $U_n \le R_n < \tau \le U_n + d_n$. In particular this is a record step. If $d_n \le B$, then $U_n \in [\tau - B, \tau)$, so

$$U_n = \tau - B + i = x + kP + i \qquad \text{for some } 0 \le i < B.$$

The choice of x gives $m_i \mid U_n$. This is allowed: U_n/L_n need not be reduced. But $m_i \mid L_n$ too, so $m_i \mid d_n = (a_n - 1)U_n - L_n$, contradicting $0 < d_n \le B < m_i$. It is the short positive jump, not the numerator value, that is impossible.

A jump may first cross several selected heights. To charge all of them to its single excess $d_n - B$, let their number be $h \ge 1$. Their spacing gives $(h-1)P < d_n$. Write $r = d_n - B \ge 1$. Since $P \ge B + 1$, $d_n = B + r \le Pr$, and hence $h \le r$. Each of those heights exceeds U_n, so monotonicity gives

$$\sum_{\substack{\tau \text{ first crossed} \\ \text{at step } n}} f(\tau) \le (d_n - B) f(U_n).$$

Each selected height above R_T has one first crossing. The heights are P apart, with the first at most $R_T + P$. For $R_N \ge R_T + P$, the intervals $[\tau, \min\{\tau + P, R_N\}]$ with $\tau \le R_N$ cover $[R_T + P, R_N]$. Monotonicity bounds each integral by $Pf(\tau)$, so the crossing bounds give

$$\sum_{\substack{T \le n < N \\ n \in \mathcal{R}}} (-V_n - B)_+ f(U_n) \ge \frac{1}{P} \int_{R_T+P}^{R_N} f(t)\, dt. \tag{11}$$

Since $R_n \to \infty$, the right side diverges for every $B \ge 1$; $B = 0$ follows by domination. Conversely, the eventual Sylvester recurrence gives by telescoping $x_n = 1/(a_n - 1)$, so $V_n = 0$ eventually. □

For example, $f(t) = 1/[t \log(et)]$ gives the sufficient condition

$$\sum_{n \in \mathcal{R}} \frac{(-V_n - B)_+}{U_n \log(eU_n)} < \infty.$$

Its finite lower bound in (11) is $P^{-1} \log(\log(eR_N)/\log(e(R_T+P)))$. Further fixed iterated logarithmic factors are allowed whenever the integral still diverges. All these weights use the same crossing theorem.

For a series, sign control alone is not enough: we bound the total change in its summands. Put $\gamma_n = a_n^2/a_{n+1} - 1$ and $\theta_n = E_n/C_n$. The defect identity gives

$$\gamma_n + \theta_n = \frac{(1 - \theta_n)(a_n - 1 + \theta_{n+1})}{a_{n+1}}, \qquad 0 < \gamma_n + \theta_n < 3/a_n$$

eventually. Positive parts do not increase absolute differences. Since $V_n = U_n \theta_n$, the summands $U_n f(U_n)(\gamma_n - B/U_n)_+$ and $(-V_n - B)_+ f(U_n)$ differ by at most $3U_n f(U_n)/a_n \le 3f(1)C_n/a_n$. This is the error *after weighting*. Now $C_n/a_n \sim qP_n/a_n^2$, since $a_n x_n \to 1$, and

$$\frac{P_{n+1}/a_{n+1}^2}{P_n/a_n^2} = \frac{a_n^3}{a_{n+1}^2} \longrightarrow 0.$$

The ratio test therefore makes the weighted error summable, even over all indices. The remainders in (9) and (10) are likewise absolutely summable: each is eventually between zero and $qP_n/a_{n+1} = O(P_n/a_n^2)$. Thus Theorem 6.1 is equivalent to finiteness of

$$\sum_{n \in \mathcal{R}} U_n f(U_n) \left(\frac{a_n^2}{a_{n+1}} - 1 - \frac{B}{U_n} \right)_+ \tag{12}$$

for some B. This is a reformulation of the extra hypothesis, not a proof of it from quadratic growth. Both series use the full jump $U_{n+1} - U_n$, including recovery from earlier decreases. Replacing it by $R_{n+1} - R_n$, or merely proving that individual summands tend to zero, would not establish the required finiteness.

Corollary 6.2 (a bound using the least common multiple)**.** *Assume the hypotheses of Problem 1.1. Write $A_n = \operatorname{lcm}(a_1, \dots, a_{n-1})$ with $A_1 = 1$. If*

$$\limsup_{n\to\infty} \frac{A_n}{a_n}\left(\frac{a_n^2}{a_{n+1}} - 1\right) < \infty,$$

then the sequence is eventually Sylvester.

Proof. Since $L_n/A_n = q/\gcd(q, A_n)$ lies in $[1, q]$, an upper bound on the expression in the corollary gives an upper bound on $(L_n/a_n)(a_n^2/a_{n+1} - 1)$; negative terms remain negative. Equation (10) therefore bounds the negative part of V_n. Choose an integer B above that eventual bound. The record series then vanishes after a finite prefix, so Theorem 6.1 applies. □

The expression in Erdős and Straus's Theorem 3 is

$$\frac{[a_1, \dots, a_k]}{a_{k+1}}\left(\frac{a_{k+1}^2}{a_{k+2}} - 1\right),$$

with a nonpositive upper limit [1, p. 132]. Setting $n = k + 1$ gives exactly $(A_n/a_n)\gamma_n$: the prefix ends before a_n, and the growth ratio begins at a_n. Tijdeman and Yuan extend this type of criterion to positive numerators [4]. Here any finite upper bound suffices under the quadratic-limit assumption. Since $A_n \mid P_n$, the LCM hypothesis is no stronger than the product hypothesis in Corollary 4.9: if the product expression is at most B, the LCM expression is at most $\max(B, 0)$. When the earlier terms are pairwise coprime the two weights agree; repeated prime factors can make the LCM much smaller. This comparison does not assert the existence of a non-Sylvester rational example satisfying one bound but not the other.

The finite upper limit means an eventual upper bound; it does not require q to divide the LCM of the preceding denominators. For positive integer summand numerators b_n, the modified error is $V_n = b_n L_n - (a_n - 1)U_n$. The companion, Section 6 under "Integer coefficients", separately proves boundedness from a lower bound on V_n and eventual constancy when $V_n/U_n \to 0$ is also assumed. The latter conclusion must not be read into the first hypothesis alone. The same section links finite examples separating them; compare Badea [3, p. 316] and Tijdeman–Yuan [4]. The companion's Section 3 compares Duverney's signed-series criterion [2, Corollary 3.2, p. 287], whose printed condition is one-sided, with the stronger absolute-convergence specialisation used there. It also records the different growth hypotheses for irrationality exponents in [14, Theorem 1, author-version p. 2]. Neither comparison is an input to the crossing theorem.

7 Further questions

Proposition 7.1 (necessary profile)**.** *The integer tail of a sequence satisfying Problem 1.1's hypotheses but not its conclusion has $E_n \neq 0$ eventually, $|E_n|/C_n \to 0$, unbounded negative magnitudes along negative indices, and*

$$\sum_n \frac{(-E_n)_+}{C_n} = \infty.$$

Proof. Absorption excludes late zeros, descent excludes an eventually nonnegative error, and Theorems 4.1 and 5.1 exclude the two finiteness conditions. □

For $B \geq 0$ put

$$F_B(X) = \sum_{\substack{n \in \mathcal{R} \\ U_n \leq X}} (-V_n - B)_+.$$

We count record steps by their *starting numerator* U_n, not by time or the running maximum. Every selected height at most X is first crossed by a jump starting below X, even if it lands above X. Thus a non-Sylvester orbit satisfies $F_B(X) \geq X/P_B - O_B(1)$, with an orbit-dependent CRT modulus P_B; $U_n \to \infty$ also makes $F_B(X)$ finite. It would therefore suffice to prove from growth and rationality that, for some fixed integer $B \geq 0$,

$$\liminf_{X \to \infty} \frac{F_B(X)}{X} = 0. \tag{13}$$

The next lemma constructs an admissible weight from this vanishing lower ratio: enumerate records by j and put $u_j = U_n$, $w_j = (-V_n - B)_+$. The resulting weighted series is finite, contradicting Theorem 6.1. The estimate remains unproved; the lemma changes what one must estimate, not the hypotheses available.

Lemma 7.2 (weights and linear density)**.** *Let u_j be positive integers, let $w_j \geq 0$, and put $F(X) = \sum_{u_j \leq X} w_j$, with the sum allowed a priori to be $+\infty$. Then $\liminf_{X\to\infty} F(X)/X = 0$ if and only if there is a finite nonincreasing $f : [1, \infty) \to [0, \infty)$ with $\int_1^\infty f(t)\,dt = \infty$ and $\sum_j w_j f(u_j) < \infty$.*

Proof. Suppose the lower limit vanishes. Since F is constant on $[m, m+1)$, choose integers $X_1 < X_2 < \cdots$ with $X_k \geq 2^k$ and $F(X_k) \leq 2^{-k} X_k$. Set

$$f(t) = \sum_{k \geq 1} \frac{\mathbf{1}_{[1, X_k]}(t)}{X_k}.$$

This function is nonincreasing and bounded by $\sum_k 2^{-k}$. Each cutoff contributes $1 - 1/X_k$ to its integral, but at most 2^{-k} to the weighted sum. Interchanging nonnegative sums therefore gives

$$\int_1^\infty f(t)\,dt = \sum_k (1 - 1/X_k) = \infty, \qquad \sum_j w_j f(u_j) = \sum_k \frac{F(X_k)}{X_k} \leq 1.$$

Conversely, suppose f is such a weight and $F(m) \geq cm$ for some $c > 0$ and all integers $m \geq M$. Monotonicity and the divergent integral force $f(X) > 0$, so $F(X) \leq$

$f(X)^{-1}\sum_j w_j f(u_j) < \infty$. Thus partial summation applies; the nonnegative differences $f(m) - f(m+1)$ preserve the lower bound on F. For $N > M$,

$$\sum_{u_j \le N} w_j f(u_j) = F(N)f(N) + \sum_{m=1}^{N-1} F(m)(f(m) - f(m+1)) \ge c\Big(Mf(M) + \sum_{m=M+1}^{N} f(m)\Big),$$

every term being nonnegative. The right side diverges as $N \to \infty$, because $\sum_{m\ge 1} f(m) \ge \int_1^\infty f(t)\,dt = \infty$. □

7.1 Arithmetic compatibility

The remaining estimate requires the denominator recurrence as well as control of the error. For example, $C_n = n^2 + 1$, $E_n = -(2n+1)$ has vanishing relative error and the necessary scalar pattern. But at $n = 2, 3, 4$ its numerators are $5, 10, 17$. The update $10 = 5a_2 - D_2$ forces $5 \mid D_2$, hence $5 \mid D_3 = a_2 D_2$; the next update would force $5 \mid 17$. Thus these numerators cannot satisfy (1). The companion's Section 14 also constructs sequences avoiding sparse fixed moduli and distinguishes that avoidance from compatibility with all later denominators. Neither property alone supplies (13).

Conversely, suppose integer sequences indexed from zero satisfy (1), with $a_n > 1$, $C_n > 0$, $D_n \ge 0$ and $E_n/C_n \to 0$. Section 4 of the companion proves that $D_n > 0$, $C_n/D_n \to 0$, $\sum_n 1/a_n = C_0/D_0$ and $a_{n+1}/a_n^2 \to 1$. The vanishing remainder $C_n/D_n \to 0$ turns the finite reciprocal identities into an infinite series. A scalar profile or finite prefix does not supply it.

Problem 7.3 (growth of repeated denominator factors)**.** For every rational-tail orbit satisfying the hypotheses but not the conclusion of Problem 1.1, must

$$\limsup_{n\to\infty} \frac{\log M_n}{n} > 0?$$

Equivalently, must there be a $K \ge 1$ for which

$$2^n \le M_n^K$$

at infinitely many indices?

Section 2 gives $C_{n+1}/C_n \to 1$; averaging logarithms therefore gives $\log C_n = o(n)$. Since $1 \le M_n \le C_n$, every such orbit already satisfies $\log M_n/n \to 0$. A positive answer therefore holds exactly when no such orbit exists: proving it from failure of Sylvester behaviour would exclude every counterexample.

A A factorial modulus for integral recursion

A constant-negative error $E_n = -m$ with $C_0 = c$ gives $C_n = c + nm$ and the identity

$$D_n + m = (a_n - 1)C_n. \tag{14}$$

In the case $m = c = 1$, where (14) reads $D_n + 1 = (a_n - 1)(n+1)$, each multiplier is determined by its predecessor. The recursion is $a_{n+1} = \mathrm{num}(n, a_n)/(n+2)$, where

$$\mathrm{num}(n, a) = (n+1)a^2 - (n+2)a + (n+3),$$

the forced numerator. We stop at the first nonintegral quotient. This procedure checks divisibility only; it does not impose the positivity conditions required of an infinite reciprocal-tail sequence. The formal exact-division condition requires exact division at each step.

Example A.1 (a nonintegral second step)**.** Starting from $a_0 = 3$, the first numerator $\mathrm{num}(0,3) = 6$ is divisible by 2, giving $a_1 = 3$. The next numerator $\mathrm{num}(1,3) = 13$ is not divisible by 3, so the recursion stops.

Direct iteration can produce very large intermediate integers. To decide whether the first h divisions are exact, however, it suffices to know the initial value modulo $(h+1)!$. Computing a pseudo-greedy orbit through residues modulo a shrinking product modulus is Koizumi's method [10, Remark 2 and Algorithm 1, pp. 13–14]; for the forced numerator that product is a factorial.

Theorem A.2 (factorial residue reduction)**.** *Let h be a nonnegative integer and let a, b be integers with $a \equiv b \pmod{(h+1)!}$. The first h steps of the recursion $a_{n+1} = \mathrm{num}(n, a_n)/(n+2)$, starting at index zero, are integral for $a_0 = a$ if and only if they are integral for $a_0 = b$.*

Proof. Division by $i+2$ consumes that factor of the modulus; no modular inverse is required. Define $M(0,i) = 1$ and $M(h+1,i) = (i+2)M(h,i+1)$. Induct on h, allowing any starting index i: inputs congruent modulo $M(h,i)$ either both complete h integral steps or neither does. The case $h = 0$ is immediate.

If $a \equiv b \pmod{(i+2)M(h,i+1)}$, the integer polynomial $\mathrm{num}(i,\cdot)$ preserves this congruence. Its values are therefore either both divisible by $i+2$ or neither is. In the latter case both recursions stop. In the former, exact division leaves the quotients congruent modulo $M(h,i+1)$, and induction supplies the remaining h steps. Finally $M(h,0) = (h+1)!$. □

For $h = 1$, the condition is $2 \mid a^2 - 2a + 3$, or equivalently that a is odd. This illustrates Theorem A.2 with modulus $2! = 2$.

The enumeration in Example 9.1 of the companion, over $2 \le a_0 < 5000$, gives at most 17 successful updates (18 values including a_0). The seed $a_0 = 1$ is excluded because it is fixed and violates $a_n \ge 2$. This finite calculation establishes only the stated range. The infinite constant-negative exclusion requires the separate argument under the hypotheses given there.

B Sources and related work

The companion *Reciprocal Sums and the Sylvester Recurrence: Further Results and Proofs* supplies full proofs and fixed-revision links: Section 2 for regular rates and cubic disagreement; Sections 4–5 for integer tails; Section 12 for the gcd and bounded-increment arguments; Section 6 for LCM results; Section 13 for scalar summability; and Section 14 for open estimates and failed constructions. A final map records dependencies. The auxiliary constructions satisfy only their stated hypotheses, not necessarily those of a rational reciprocal series.

For the earlier integer-remainder method see Erdős–Straus [12, Theorem 2.1, pp. 85–86]. The companion's Section 3 distinguishes polynomial Cantor series [13, Theorem 2.2, pp. 39–40] and the freely selectable series of Problem #270 [15, Theorems 1–2] from the present setting. The Isabelle/HOL work [16] formalises older Erdős–Straus criteria, not Problem #243. General-rate extraction uses the Gamma-ratio asymptotics in [17]; the cubic case uses the explicit F_n.

Indexing of the canonical tails. Section 2 starts with $C_1 = p$, $D_1 = q$. Setting $a_0 = 1$, $D_0 = q$, $C_0 = p + q$ extends the recurrences by one update, but gives neither $a_0 > 1$ nor strict increase at that step. The zero-based formal cubic theorem instead uses N with $a_N > N$, $b_n = n + 1$ for $0 \le n < N$ and $b_n = a_n$ thereafter. This sequence is strictly increasing and changes the reciprocal sum by a rational finite sum without shifting n or $3/n$ on the tail. See the companion's indexing subsection.

C Verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The links identify fixed-revision Lean declarations and recorded Comparator checks. The printed cubic argument uses Chebotarev; the formal square-specialisation proof uses a Dedekind-zeta pole comparison. Their common conclusion does not check each step of the printed proof. This revision includes no new Lean build, Comparator replay or independent review. The evidence record retains source-pin and index gaps, and the estimate in Section 7 remains open.
The formal factorial residue reduction uses the shrinking-modulus induction, polynomial congruence, and cancellation after exact division. Its modulus is identified by the ascending-factorial formula and its factorial value at the initial index.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.2	Lean · Comparator	Proposition 3.1	Lean · Comparator
Theorem 4.1	Lean · Comparator	Proposition 4.2	Lean · Comparator
Theorem 4.3	Lean · Comparator	Corollary 4.4	Lean · Comparator
Proposition 4.5	Lean · Comparator	Lemma 4.6	Lean · Comparator
Theorem 4.7	Lean · Comparator	Corollary 4.8	Lean · Comparator
Corollary 4.9	Lean · Comparator	Corollary 4.10	Lean · Comparator
Theorem 5.1	Lean · Comparator	Theorem 6.1	Lean · Comparator
Corollary 6.2	Lean · Comparator	Proposition 7.1	Lean · Comparator
Lemma 7.2	Lean · Comparator	Theorem A.2	Lean · Comparator